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Type 'help;' or '\h' for help. Type '\c' to clear the current input statement.

```
mysql> USE EmployeeDB;
```

Database changed

```
mysql> DROP TABLE IF EXISTS DBMS_Stud;
```

Query OK, 0 rows affected, 1 warning (0.05 sec)

```
mysql> CREATE TABLE DBMS_Stud
```

```
-> (
```

```
->     RegNo INT AUTO_INCREMENT PRIMARY KEY,
```

```
->     StudentName VARCHAR(30),
```

```
->     Gender CHAR(1),
```

```
->     City VARCHAR(20)
```

```
-> );
```

Query OK, 0 rows affected (0.05 sec)

```
mysql> DESCRIBE DBMS_Stud;
```

Field	Type	Null	Key	Default	Extra
RegNo	int	NO	PRI	NULL	auto_increment
StudentName	varchar(30)	YES		NULL	
Gender	char(1)	YES		NULL	
City	varchar(20)	YES		NULL	

4 rows in set (0.01 sec)

```
mysql> INSERT INTO DBMS_Stud(StudentName, Gender, City)
```

```
-> VALUES
```

```
-> ('Rahul','M','Chennai'),
```

```
-> ('Priya','F','Hyderabad'),
```

```
-> ('Kamal','M','Bangalore'),
```

```
-> ('Anitha','F','Coimbatore');
```

Query OK, 4 rows affected (0.07 sec)

Records: 4 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM DBMS_Stud;
```

RegNo	StudentName	Gender	City
1	Rahul	M	Chennai
2	Priya	F	Hyderabad
3	Kamal	M	Bangalore
4	Anitha	F	Coimbatore

4 rows in set (0.00 sec)

```
mysql> INSERT INTO DBMS_Stud
-> (RegNo, StudentName, Gender, City)
-> VALUES
-> (100, 'Ramesh', 'M', 'Madurai');
Query OK, 1 row affected (0.06 sec)
```

```
mysql> SELECT * FROM DBMS_Stud;
+-----+-----+-----+-----+
| RegNo | StudentName | Gender | City      |
+-----+-----+-----+-----+
|      1 | Rahul       | M      | Chennai   |
|      2 | Priya       | F      | Hyderabad |
|      3 | Kamal       | M      | Bangalore |
|      4 | Anitha      | F      | Coimbatore |
|     100 | Ramesh      | M      | Madurai   |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

Exp12

```
mysql> USE EmployeeDB;
Database changed
mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION While_Loop()
-> RETURNS VARCHAR(100)
-> DETERMINISTIC
-> BEGIN
->     DECLARE x INT DEFAULT 1;
->     DECLARE str VARCHAR(100) DEFAULT '';
->
->     WHILE x <= 5 DO
->         SET str = CONCAT(str, x, ' ');
->         SET x = x + 1;
->     END WHILE;
->
->     RETURN str;
-> END$$
Query OK, 0 rows affected (0.11 sec)
```

```
mysql>
mysql> DELIMITER ;
mysql> SELECT While_Loop();
+-----+
| While_Loop() |
+-----+
| 1 2 3 4 5    |
+-----+
1 row in set (0.03 sec)
```

```
mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION Repeat_Loop()
-> RETURNS INT
-> DETERMINISTIC
-> BEGIN
->     DECLARE income INT DEFAULT 1000;
->
->     REPEAT
->         SET income = income + 500;
->     UNTIL income >= 4000
->     END REPEAT;
->
->     RETURN income;
-> END$$
Query OK, 0 rows affected (0.06 sec)
```

```
mysql>
mysql> DELIMITER ;
mysql> SELECT Repeat_Loop();
+-----+
| Repeat_Loop() |
+-----+
|           4000 |
+-----+
1 row in set (0.00 sec)
```

Exp13

```
mysql> USE EmployeeDB;
Database changed
mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION Income_Level(monthly_value INT)
-> RETURNS VARCHAR(20)
-> DETERMINISTIC
-> BEGIN
->     DECLARE income_level VARCHAR(20);
->
->     CASE
->         WHEN monthly_value <= 4000 THEN
->             SET income_level = 'Low Income';
->
->         WHEN monthly_value <= 5000 THEN
->             SET income_level = 'Avg Income';
->
->         ELSE
->             SET income_level = 'High Income';
->     END CASE;
->
->     RETURN income_level;
-> END$$
```

Query OK, 0 rows affected (0.06 sec)

```
mysql>
mysql> SELECT Income_Level(3500);
-> SELECT Income_Level(4500);
-> SELECT Income_Level(7000);
-> DELIMITER $$
```

```
+-----+
| Income_Level(3500) |
+-----+
| Low Income        |
+-----+
1 row in set (0.00 sec)
```

```
+-----+
| Income_Level(4500) |
+-----+
| Avg Income         |
+-----+
1 row in set (0.01 sec)
```

```
+-----+
| Income_Level(7000) |
+-----+
| High Income        |
+-----+
1 row in set (0.02 sec)
```

```

mysql> CREATE FUNCTION Loop_Income()
-> RETURNS INT
-> DETERMINISTIC
-> BEGIN
->     DECLARE income INT DEFAULT 1000;
->
->     myloop: LOOP
->
->         SET income = income + 500;
->
->         IF income < 4000 THEN
->             ITERATE myloop;
->         END IF;
->
->         LEAVE myloop;
->
->     END LOOP;
->
->     RETURN income;
-> END$$

```

Query OK, 0 rows affected (0.06 sec)

```

mysql>
mysql> DELIMITER ;
mysql> DELIMITER $$
mysql>
mysql> CREATE FUNCTION Loop_Income()
-> RETURNS INT
-> DETERMINISTIC
-> BEGIN
->     DECLARE income INT DEFAULT 1000;
->
->     myloop: LOOP
->
->         SET income = income + 500;
->
->         IF income < 4000 THEN
->             ITERATE myloop;
->         END IF;
->
->         LEAVE myloop;
->
->     END LOOP;
->
->     RETURN income;
-> END$$

```

ERROR 1304 (42000): FUNCTION Loop_Income already exists

```

mysql>
mysql> DELIMITER ;
mysql> SELECT Loop_Income();
+-----+
| Loop_Income() |
+-----+
|           4000 |

```

Exp 14 mysql> CREATE TABLE CLASS

```
-> (  
->     Name VARCHAR(20),  
->     ID INT PRIMARY KEY  
-> );
```

Query OK, 0 rows affected (0.08 sec)

mysql> DESCRIBE CLASS;

```
+-----+-----+-----+-----+-----+-----+  
| Field | Type          | Null | Key | Default | Extra |  
+-----+-----+-----+-----+-----+-----+  
| Name  | varchar(20)   | YES  |     | NULL    |       |  
| ID    | int           | NO   | PRI | NULL    |       |  
+-----+-----+-----+-----+-----+-----+  
2 rows in set (0.00 sec)
```

mysql> INSERT INTO CLASS VALUES

```
-> ('Arun',1),  
-> ('Bharath',2),  
-> ('Charles',3),  
-> ('David',4),  
-> ('Alpha',5);
```

Query OK, 5 rows affected (0.06 sec)

Records: 5 Duplicates: 0 Warnings: 0

mysql> SELECT * FROM CLASS;

```
+-----+-----+  
| Name  | ID |  
+-----+-----+  
| Arun  | 1 |  
| Bharath | 2 |  
| Charles | 3 |  
| David  | 4 |  
| Alpha  | 5 |  
+-----+-----+
```

5 rows in set (0.00 sec)

mysql> START TRANSACTION;

Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE CLASS

```
-> SET Name='Bravo'  
-> WHERE ID=5;
```

Query OK, 1 row affected (0.00 sec)

Rows matched: 1 Changed: 1 Warnings: 0

mysql> SELECT * FROM CLASS;

```
+-----+-----+  
| Name  | ID |  
+-----+-----+  
| Arun  | 1 |  
| Bharath | 2 |  
| Charles | 3 |  
| David  | 4 |  
| Bravo  | 5 |  
+-----+-----+
```

Exp 16

```
mysql> INSERT INTO student_info VALUES
-> (101, 'Rahul', 'M', 'Chennai'),
-> (102, 'Priya', 'F', 'Hyderabad'),
-> (103, 'Kamal', 'M', 'Bangalore'),
-> (104, 'Anitha', 'F', 'Coimbatore'),
-> (105, 'Ramesh', 'M', 'Madurai');
```

Query OK, 5 rows affected (0.07 sec)
Records: 5 Duplicates: 0 Warnings: 0

```
mysql> SELECT * FROM student_info;
+-----+-----+-----+-----+
| RegNo | Student_Name | Gender | City      |
+-----+-----+-----+-----+
| 101   | Rahul        | M      | Chennai   |
| 102   | Priya        | F      | Hyderabad |
| 103   | Kamal        | M      | Bangalore |
| 104   | Anitha       | F      | Coimbatore |
| 105   | Ramesh       | M      | Madurai   |
+-----+-----+-----+-----+
5 rows in set (0.00 sec)
```

```
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE GetCustomerLevel(
->     IN p_customernumber INT,
->     OUT p_customerlevel VARCHAR(20)
-> )
-> BEGIN
->     DECLARE v_creditlimit DECIMAL(10,2);
->
->     SELECT CreditLimit
->     INTO v_creditlimit
->     FROM customers
->     WHERE CustomerNumber = p_customernumber;
->
->     IF v_creditlimit >= 100000 THEN
->         SET p_customerlevel = 'PLATINUM';
->
->     ELSEIF v_creditlimit >= 50000 THEN
->         SET p_customerlevel = 'GOLD';
->
->     ELSE
->         SET p_customerlevel = 'SILVER';
->     END IF;
-> END$$
Query OK, 0 rows affected (0.07 sec)
```

Exp 17

```
mysql> CREATE FUNCTION GetCustomerLevel(p_customer_number INT)
-> RETURNS VARCHAR(20)
-> DETERMINISTIC
-> BEGIN
->     DECLARE v_credit_limit DECIMAL(10,2);
->     DECLARE customer_level VARCHAR(20);
->
->     SELECT CreditLimit
->     INTO v_credit_limit
->     FROM customers
->     WHERE CustomerNumber = p_customer_number;
->
->     IF v_credit_limit > 50000 THEN
->         SET customer_level = 'PLATINUM';
->
->     ELSEIF v_credit_limit >= 10000
->         AND v_credit_limit <= 50000 THEN
->         SET customer_level = 'GOLD';
->
->     ELSE
->         SET customer_level = 'SILVER';
->     END IF;
->
->     RETURN customer_level;
->
```

```
mysql> SELECT * FROM customers;
```

CustomerNumber	CustomerName	CreditLimit
101	Arun	75000.00
102	Priya	30000.00
103	Kamal	5000.00
104	Anitha	120000.00
105	Ramesh	15000.00


```

5 rows in set (0.00 sec)END$$
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE Factorial(
->     IN n INT,
->     OUT result BIGINT
-> )
-> BEGIN
->
->     DECLARE temp_result BIGINT;
->
->     IF n <= 1 THEN
->         SET result = 1;
->
->     ELSE
->         CALL Factorial(n - 1, temp_result);
->         SET result = n * temp_result;
->     END IF;
->
-> END$$
ERROR 1304 (42000): PROCEDURE Factorial already exists
mysql>
mysql> DELIMITER ;
mysql> CALL Factorial(5, @result);
Query OK, 0 rows affected (0.00 sec)

mysql>
mysql> SELECT @result AS Factorial;
+-----+
| Factorial |
+-----+
|      120 |
+-----+
1 row in Records: 3  Duplicates: 0  Warnings: 0

```

```

Exp 18
mysql> DELIMITER $$
mysql>
mysql> CREATE PROCEDURE build_email_list
-> (
->     INOUT email_list VARCHAR(4000)
-> )
-> BEGIN
->
->     DECLARE v_finished INTEGER DEFAULT 0;
->     DECLARE v_email VARCHAR(100) DEFAULT '';
->
->     DECLARE email_cursor CURSOR FOR
->         SELECT email FROM employees;
->
->     DECLARE CONTINUE HANDLER FOR NOT FOUND
->         SET v_finished = 1;
->
->     OPEN email_cursor;
->
->     get_email: LOOP
->
->         FETCH email_cursor INTO v_email;
->
->         IF v_finished = 1 THEN
->             LEAVE get_email;
->         END IF;
->
->         SET email_list = CONCAT(email_list, v_email, ';');
->
->     END LOOP get_email;
->
->     CLOSE email_cursor;
->
-> END$$

```

Query OK, 0 rows affected (0.08 sec)

```

mysql>
mysql> DELIMITER ;
mysql> SET @email_list = '';
Query OK, 0 rows affected (0.00 sec)

```

```

mysql> CALL build_email_list(@email_list);
Query OK, 0 rows affected (0.05 sec)

```

```

mysql> SELECT @email_list AS Email_List;
+-----+
| Email_List |
+-----+
| pharry@warnerbros.com;kclark@dccomics.com;stony@marvel.com; |
+-----+
1 row in set (0.00 sec) set (0.00 sec)

```

Exp 19

```
mysql> DELIMITER $$
```

```
mysql> CREATE TRIGGER student_before_update
```

```
-> BEFORE UPDATE ON STUDENT
```

```
-> FOR EACH ROW
```

```
-> BEGIN
```

```
-> INSERT INTO EMPLOYEES_AUDIT
```

```
-> (ID, Name, Email)
```

```
-> VALUES
```

```
-> (OLD.ID, OLD.Name, OLD.Email);
```

```
-> END$$
```

Query OK, 0 rows affected (0.02 sec)

```
mysql> DELIMITER ;
```

```
mysql> UPDATE STUDENT
```

```
-> SET Email = 'rahul2026@gmail.com'
```

```
-> WHERE ID = 1;
```

Query OK, 1 row affected (0.02 sec)

Rows matched: 1 Changed: 1 Warnings: 0

```
mysql> SELECT * FROM STUDENT;
```

```
+----+-----+-----+
```

```
| ID | Name | Email      |
```

```
+----+-----+-----+
```

```
| 1 | Rahul | rahul2026@gmail.com |
```

```
| 2 | Priya | priya@gmail.com    |
```

```
| 3 | Kamal | kamal@gmail.com    |
```

```
+----+-----+-----+
```

3 rows in set (0.00 sec)

